Assignment No.: 02

1: Pull and Run Image

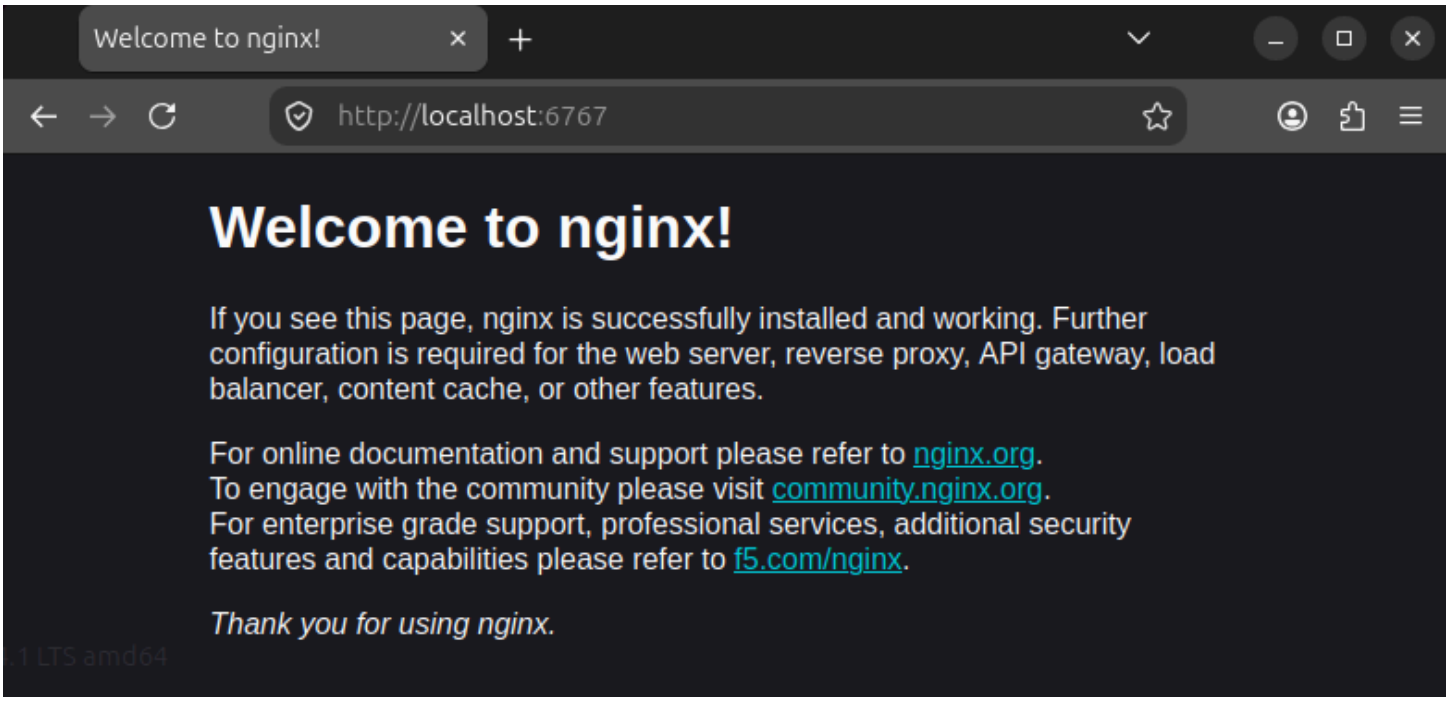
1. Pull the nginx image from Docker Hub.

```
student@student-VMware-Virtual-Platform:~$ docker pull nginx:latest
latest: Pulling from library/nginx
Digest: sha256:4c41e2d0a3a55b325ed572cd0200bdb72af1433c3daa2bca6ef8eea68ce504a2
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest
```

2. Run a container from the image.

```
student@student-VMware-Virtual-Platform:~$ docker run --name my-nginx -d -p 6767:80 nginx
488fe470cfcd53bdf6875c5e3748753aa3f7a4cf9d1e9336cefd812fafb1afa1
```

3. Check running containers.



4. Stop the container.

5. Remove the container.

```
student@student-VMware-Virtual-Platform:~$ docker container ls
CONTAINER ID   IMAGE     COMMAND                  CREATED          STATUS          PORTS                                      NAMES
488fe470cfcd   nginx     "/docker-entrypoint.…"   10 minutes ago   Up 10 minutes   0.0.0.0:6767->80/tcp, [::]:6767->80/tcp   my-nginx
student@student-VMware-Virtual-Platform:~$ docker stop 48
48
student@student-VMware-Virtual-Platform:~$ docker rm 48
48
student@student-VMware-Virtual-Platform:~$ docker container ls
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
student@student-VMware-Virtual-Platform:~$
```

2: Image Commands Practice

1. Pull alpine image.

```
student@student-VMware-Virtual-Platform:~$ docker pull alpine
Using default tag: latest
latest: Pulling from library/alpine
7b229517ed22: Pull complete
3da1513d2124: Download complete
3ec36f3670df: Download complete
Digest: sha256:be2656061b536c1883c4034790e21d0a09c5aaaf51968b89368a3c0371ea2c9a
Status: Downloaded newer image for alpine:latest
docker.io/library/alpine:latest
```

2. List all images available locally.

```
student@student-VMware-Virtual-Platform:~$ docker images

IMAGE                                                                                 ID             DISK USAGE   CONTENT SIZE
alpine:latest                                                                         be2656061b53       13MB         3.94MB
busybox:latest                                                                        fd7dc98638c8     6.81MB         2.23MB
dock_compose-app:latest                                                               1ff0a32c3564      187MB         45.6MB
docker-compose-demo-app:latest                                                        c9e4512d7168      287MB         70.6MB
dockerid/custom-nginx:1.0                                                             323bd910caf2      240MB         63.9MB
harsh1628/custom-nginx:1.0                                                            338bce652453      240MB         63.9MB
harsh1628/custom-nx:1.0                                                               f5c7c8c03589      240MB         63.9MB
harsh1628/my-c:1.0                                                                    042bbbdc3f9b      2.2GB          562MB
hello-world-python:latest                                                             8c77bec4e44f      176MB         44.1MB
httpd:latest                                                                          a30cd4cd57f3      177MB         47.7MB
my-c:latest                                                                           042bbbdc3f9b      2.2GB          562MB
mysql:8                                                                               76f0358884cb     1.12GB          254MB
mysql:latest                                                                          f51c9c01c16b     1.08GB          304MB
newdelthis/custom-nginx:1.0@sha256:fc1e7ec7359701ba74127e421d0f578fcf35058fa900fe43f8cc2659e341d0d2  fc1e7ec73597      279MB         72.9MB
nginx:latest                                                                          4c41e2d0a3a5      243MB         66.8MB
phpmyadmin/phpmyadmin:latest                                                          42a200db07b4     1.09GB          221MB
```

3. Inspect the alpine image.

```
student@student-VMware-Virtual-Platform:~$ docker inspect alpine
[
    {
        "Id": "sha256:be2656061b536c1883c4034790e21d0a09c5aaaf51968b89368a3c0371ea2c9a",
        "RepoTags": [
            "alpine:latest"
        ],
        "RepoDigests": [
            "alpine@sha256:be2656061b536c1883c4034790e21d0a09c5aaaf51968b89368a3c0371ea2c9a"
        ],
        "Comment": "buildkit.dockerfile.v0",
        "Created": "2026-10-21T22:37:22.870489069Z",
        "Config": {
            "Env": [
                "PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin"
            ],
            "Cmd": [
                "/bin/sh"
```

4. Remove the alpine image.

```
student@student-VMware-Virtual-Platform:~$ docker rmi alpine
Untagged: alpine:latest
Deleted: sha256:be2656061b536c1883c4034790e21d0a09c5aaaf51968b89368a3c0371ea2c9a
```

5. What happens if you try to remove a running container's image? Explain.

-> Docker **safely blocks the operation and refuses to delete the image** because running containers rely directly on its read-only layers to function. Instead of deleting the files, the Docker daemon throws a conflict error to prevent the active container from breaking.

Error response from daemon: conflict: unable to delete <image_id> (must be forced) - image is being used by running container <container_id>

To bypass this, you must either stop and remove the container first, or use the force flag (docker rmi -f), which merely untags the image while keeping its data intact until the container exits.

3: Container Commands

1. Run a container using ubuntu image in interactive mode.

```
student@student-VMware-Virtual-Platform:~$ docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
d9d511d19a19: Pull complete
435bf5c20694: Pull complete
2e7e55092090: Download complete
Digest: sha256:c098726dd9b9733f3e885f8de0ef375206a311b0a317eadc40d347b1115fb944
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest
```

```
student@student-VMware-Virtual-Platform:~$ docker run -dit --name my-ubu ubuntu
71cd78cf019f8afeb5d767d8eaaf7a99fdce9942d0f094a795b998b035d63466
student@student-VMware-Virtual-Platform:~$ docker ps
CONTAINER ID   IMAGE     COMMAND       CREATED         STATUS         PORTS     NAMES
71cd78cf019f   ubuntu    "/bin/bash"   7 seconds ago   Up 7 seconds             my-ubu
student@student-VMware-Virtual-Platform:~$ docker exec -it my-ubu
```

2. Execute basic Linux command inside container (ls , pwd).

```
student@student-VMware-Virtual-Platform:~$ docker exec -it my-ubu /bin/bash
root@71cd78cf019f:/# ls
bin  boot  dev  etc  home  lib  lib64  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
root@71cd78cf019f:/# pwd
/
root@71cd78cf019f:/#
```

3. Exit the container.

4. List all containers (including stopped).

5. Restart the stopped container.

```
student@student-VMware-Virtual-Platform:~$ docker ps -a
CONTAINER ID   IMAGE     COMMAND       CREATED         STATUS         PORTS     NAMES
71cd78cf019f   ubuntu    "/bin/bash"   3 minutes ago   Up 3 minutes             my-ubu
student@student-VMware-Virtual-Platform:~$ docker stop my-ubu
my-ubu
student@student-VMware-Virtual-Platform:~$ docker restart my-ubu
my-ubu
student@student-VMware-Virtual-Platform:~$ docker ps -a
CONTAINER ID   IMAGE     COMMAND       CREATED         STATUS         PORTS     NAMES
71cd78cf019f   ubuntu    "/bin/bash"   4 minutes ago   Up 4 seconds             my-ubu
```

4: Naming and Renaming

1. Run an nginx container with name web_server .
2. Verify the container name.

```
student@student-VMware-Virtual-Platform:~$ docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
Digest: sha256:4c41e2d0a3a55b325ed572cd0200bdb72af1433c3daa2bca6ef8eea68ce504a2
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest
student@student-VMware-Virtual-Platform:~$ docker run --name web_server -d -p 6767:80 nginx
63aeae02c6284c5cc2ce929f37a989170f83a29823ef0eab8dc9758952bce470
student@student-VMware-Virtual-Platform:~$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS                                      NAMES
63aeae02c628   nginx     "/docker-entrypoint.…"   7 seconds ago   Up 6 seconds   0.0.0.0:6767->80/tcp, [::]:6767->80/tcp    web_server
71cd78cf019f   ubuntu    "/bin/bash"              7 minutes ago   Up 3 minutes                                              my-ubu
```

3. Stop the container.
4. Rename the container to my_server .

```
student@student-VMware-Virtual-Platform:~$ docker stop web_server
web_server
student@student-VMware-Virtual-Platform:~$ docker rename web_server my_server
student@student-VMware-Virtual-Platform:~$ docker ps
CONTAINER ID   IMAGE     COMMAND       CREATED         STATUS         PORTS     NAMES
71cd78cf019f   ubuntu    "/bin/bash"   9 minutes ago   Up 4 minutes             my-ubu
student@student-VMware-Virtual-Platform:~$ docker restart my
my_server  my-ubu
student@student-VMware-Virtual-Platform:~$ docker restart my_server
my_server
student@student-VMware-Virtual-Platform:~$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS                                      NAMES
63aeae02c628   nginx     "/docker-entrypoint.…"   2 minutes ago   Up 2 seconds   0.0.0.0:6767->80/tcp, [::]:6767->80/tcp    my_server
71cd78cf019f   ubuntu    "/bin/bash"              9 minutes ago   Up 4 minutes                                              my-ubu
```

5. Remove the container.

```
student@student-VMware-Virtual-Platform:~$ docker stop my_server
my_server
student@student-VMware-Virtual-Platform:~$ docker rm my_server
my_server
```

5: Tagging Docker Image

1. Pull nginx image.

```
student@student-VMware-Virtual-Platform:~$ docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
Digest: sha256:4c41e2d0a3a55b325ed572cd0200bdb72af1433c3daa2bca6ef8eea68ce504a2
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest
```

2. Tag it as mynginx:v1 .

3. Tag it again as mynginx:latest .

```
student@student-VMware-Virtual-Platform:~$ docker tag nginx mynginx:v1
student@student-VMware-Virtual-Platform:~$ docker tag nginx  mynginx:latest
```

4. List images and verify tags.

```
student@student-VMware-Virtual-Platform:~$ docker images

IMAGE                                                                      ID             DISK USAGE   CONTENT SIZE
busybox:latest                                                             fd7dc98638c8       6.81MB         2.23MB
dock_compose-app:latest                                                    1ff0a32c3564        187MB         45.6MB
docker-compose-demo-app:latest                                             c9e4512d7168        287MB         70.6MB
dockerid/custom-nginx:1.0                                                  323bd910caf2        240MB         63.9MB
harsh1628/custom-nginx:1.0                                                 338bce652453        240MB         63.9MB
harsh1628/custom-nx:1.0                                                    f5c7c8c03589        240MB         63.9MB
harsh1628/my-c:1.0                                                         042bbbdc3f9b        2.2GB          562MB
hello-world-python:latest                                                  8c77bec4e44f        176MB         44.1MB
httpd:latest                                                               a30cd4cd57f3        177MB         47.7MB
my-c:latest                                                                042bbbdc3f9b        2.2GB          562MB
mynginx:latest                                                             4c41e2d0a3a5        243MB         66.8MB
mynginx:v1                                                                 4c41e2d0a3a5        243MB         66.8MB
mysql:8                                                                    76f0358884cb        1.12GB         254MB
mysql:latest                                                               f51c9c01c16b        1.08GB         304MB
newdelthis/custom-nginx:1.0@sha256:fc1e7ec7359701ba74127e421d0f578fcf35058fa900fe43f8cc2659e341d0d2  fc1e7ec73597        279MB         72.9MB
nginx:latest                                                               4c41e2d0a3a5        243MB         66.8MB
phpmyadmin/phpmyadmin:latest                                               42a200db07b4        1.09GB         221MB
ubuntu:latest                                                              c098726dd9b9        163MB         45.9MB
```

5. Remove one tag and observe result.

```
student@student-VMware-Virtual-Platform:~$ docker rmi mynginx:latest
Untagged: mynginx:latest
```

6: Push Image to Docker Hub

1. Login to Docker Hub using docker login .

```
student@student-VMware-Virtual-Platform:~$ docker login
Authenticating with existing credentials... [Username: harsh1628]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded
```

2. Tag your image as:

```
student@student-VMware-Virtual-Platform:~$ docker tag nginx harsh1628/mynginx:v1
```

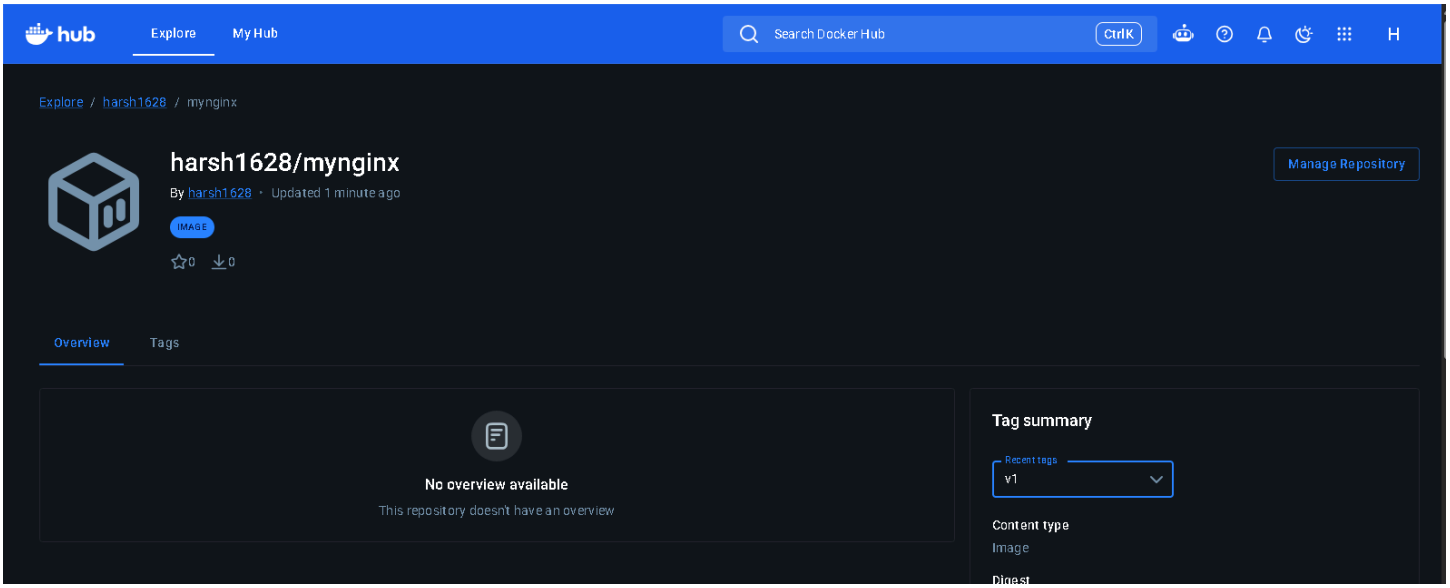
3. Push the image to Docker Hub.

4. Logout from Docker.

```
student@student-VMware-Virtual-Platform:~$ docker push harsh1628/mynginx:v1
The push refers to repository [docker.io/harsh1628/mynginx]
dd00b11a22aa: Mounted from harsh1628/custom-nginx
f2be67225471: Mounted from harsh1628/custom-nginx
d5f3a4724daf: Mounted from harsh1628/custom-nginx
9a2c3eb3312e: Mounted from harsh1628/custom-nginx
24bf03b6a474: Mounted from harsh1628/custom-nginx
a856bfc2a772: Mounted from harsh1628/custom-nginx
8334177aa35b: Mounted from harsh1628/custom-nginx
v1: digest: sha256:59a9ae22b8933b536391e6c1708fa761297e922809c97675c5a56ba316be62af size: 2290

i Info → Not all multiplatform-content is present and only the available single-platform image was pushed
        sha256:4c41e2d0a3a55b325ed572cd0200bdb72af1433c3daa2bca6ef8eea68ce504a2 -> sha256:59a9ae22b8933b536391e6c1708fa761297e922809c97675c5a56ba316be62af
student@student-VMware-Virtual-Platform:~$ docker logout
Removing login credentials for https://index.docker.io/v1/
```

5. Verify image on Docker Hub website.

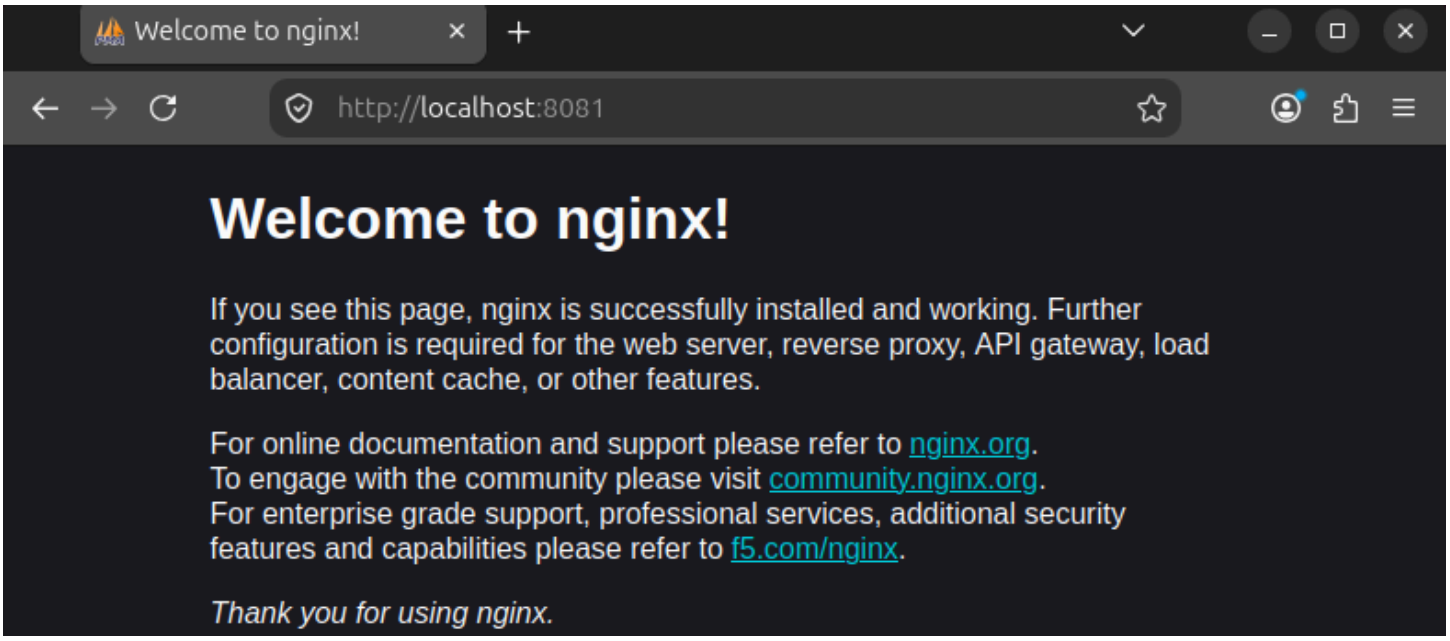


7: Basic Networking and Port Mapping

1. Run an nginx container with port mapping:

```
student@student-VMware-Virtual-Platform:~$ docker run --name my-nginx -d -p 8081:80 nginx
9bb21c39654c82f4716a7e3e4af33c76d81d92aff11ceb4da1e50470c36e4290
```

2. Open browser and access localhost:8081 .



3. List Docker networks.

```
student@student-VMware-Virtual-Platform:~$ docker network ls
NETWORK ID     NAME                           DRIVER    SCOPE
3211bc892d5e   bridge                         bridge    local
2668b1e9139d   demo_default                   bridge    local
bd4531d58fcb   docker-compose-demo_default    bridge    local
27d6fb9bd135   docker_gwbridge                bridge    local
e05cc1e1583d   host                           host      local
th71s0wahy67   ingress                        overlay   swarm
7b5974a55708   itiss                          bridge    local
93c7d9d90e84   none                           null      local
```

4. Inspect default bridge network.

```
student@student-VMware-Virtual-Platform:~$ docker network inspect 7b
[
    {
        "Name": "itiss",
        "Id": "7b5974a557082d0db894fd28750903e6a33541d907333100d86ee20ebf984760",
        "Created": "2026-10-23T23:28:55.800318019+05:30",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": {},
            "Config": [
                {
                    "Subnet": "172.18.0.0/16",
                    "Gateway": "172.18.0.1"
                }
            ]
        },
```

5. Explain what port mapping does in Docker.

Port mapping routes external network traffic into an isolated container by linking a port on your host machine to a port inside the container.

Using the -p <Host-Port>:<Container-Port> flag, it acts as a bridge through Docker's network barrier.

For example, -p 8080:80 forwards traffic from localhost:8080 to the container's internal web server on port 80. This allows you to expose applications to your local machine or internet and run multiple identical services simultaneously by assigning each to a unique host port.